

# Tirumalesh Nagothi

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## PROFESSIONAL SUMMARY

Python Backend Developer with 3+ years of experience building scalable, high-performance backend systems, cloud-native applications, and AI/LLM-powered platforms. Specializes in FastAPI, Django, microservices, and event-driven architectures using PostgreSQL, MongoDB, Redis, Kafka, Docker, and Kubernetes on AWS. Experienced delivering production-grade APIs, RAG/LangChain pipelines, and observability with Prometheus and Grafana. Open to relocation across the United States.

## TECHNICAL SKILLS

- Languages:** Python, SQL, Bash, JavaScript Backend & APIs: FastAPI, Django, Flask, REST APIs, GraphQL, SQLAlchemy, Microservices, OAuth 2.0, JWT, Celery
- Databases & Messaging:** PostgreSQL, MongoDB, Redis, Apache Kafka
- Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, NGINX
- AI & Data:** LangChain, RAG, Pinecone, LLMs, PySpark, Pandas, NumPy, ETL Pipelines
- Observability & Practices:** OpenTelemetry, Prometheus, Grafana, CloudWatch, Agile, TDD, SOLID, System Design
- Mobile:** React Native
- Certifications:** IBM Data Science, Google Data Analytics

## PROFESSIONAL EXPERIENCE

### Software Engineer | Prozech Solution | USA Feb 2026 – Present

- Built an asynchronous backend document pipeline using Python and FastAPI that converts unstructured documents into structured data, processing 5,000 documents/day without blocking core system operations.
- Implemented schema validation and confidence scoring on AI outputs, automatically routing low-confidence results to human review to maintain data integrity at scale.
- Reduced AI processing cost by 88% by routing simpler documents to lightweight models and reserving stronger models for complex cases.
- Designed front-end review interfaces for correcting and validating AI output, closing the feedback loop to raise model accuracy from 75% to 98%.

### Founding Software Engineer | Rivalarc | USA Sep 2024 – Jan. 2026

- Built a React Native mobile app from scratch, delivering core flows including faceoffs, voting, Explore feed, profiles, notifications, tournaments, and scorecards.
- Designed backend REST APIs with FastAPI, MongoDB, and Redis for users, voting, tournaments, rewards, admin tools, and session management; implemented anti-fraud voting controls cutting duplicate/fake votes by 90%+.
- Automated tournament lifecycle management (entry windows, rounds, deadlines, results) and built a media pipeline on AWS S3 and Lambda for fast in-app uploads.
- Owned cloud deployment using Docker, NGINX, and AWS load balancing, cutting deployment overhead 40%; supported launch to 1,250+ users and 500+ downloads.

### Software Engineer | Leap of Faith Technologies, Inc. | Libertyville, IL Jun 2024 – Aug. 2024

- Built data pipelines using BigQuery for MedOptiX, an AI system automating Medicare MIPS reporting, aggregating and filtering clinical data from multiple sources.
- Helped design LLM prompts that extract MIPS-relevant answers from clinical notes, cutting evaluation time per note from a manual process to under 1 minute.

### MLOps Engineer | TCR Innovation | Mumbai, India Jun 2021 – Nov 2022

- Built and tuned classification models (logistic regression, random forest, SVM) in Python for cancer diagnosis prediction, achieving 83% recall on the best model through careful data cleaning and feature engineering.

## EDUCATION

### Master of Science in Computer Science Jan 2023 – Dec 2024

Illinois Institute of Technology Chicago, IL

### Bachelor of Technology in Computer Science & Engineering Jun 2018 – Apr 2022

GITAM Deemed University India